

Task Laboratory Week4

The work for this week is to extend the environment defined in the weld bicycle assembly factory by using CArtaGo approach. **This work could be done in group, but it is recommended to try it individually first.**

- 1) On the Laboratory Folder you could find the following two libraries that are needed to accomplish this task:

```
jaca-2.5-SNAPSHOT.jar  
cartago-2.5-SNAPSHOT.jar
```

- 2) Modify on the file "factory1.mas2j" the declaration of `binagent` as follows:

```
binagent agentArchClass MixedAgentArch  
agentArchClass jaca.CAgentArch #6;
```

- 3) Create a CArtaGo bin artifact following the indications on the examples document on folder Laboratory that includes an operation "`refill_bin(N)`" whose behaviour must be equivalent to "`refillBin(N)`" defined on "`facIenv.java`" on "`FactoryModel`" class.

Note that it changes the value for a shared array that is also used by the "`pickPart(Ag,N)`" method, and that its value it is used to update the perception "`binfull(N)`" on "`binagentN`".

- 4) Create the file "`MixedAgentArch`" in a similar way to defined on the last slide on "`theJasonEnvironment.pdf`" file.

Note that you could modify the "`act(Action)`" method as you needed.

- 5) Modify the "`facIenv.java`" file including the "`import jaca.CartagoEnvironment;`" and the "`private CartagoEnvironment cartagoEnv;`" and "`startCartago(args)`" lines in right places. Modify also the "`updatePercepts()`" method to comment or remove the actualization of perception for "`binfull(N)`" belief that it is not needed anymore because it is provided directly from CArtaGo artifact.